MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
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DEPARTMENT OF AERONAUTICAL & AUTOMOBILE ENGINEERING			
END SEMESTER EXAMINATION		**NOV, 2024**	
Level	**B.Tech**	**Curriculum**	**2018**
Program Name	**Aeronautical Engineering**	**Course Type**	**PE**
Course Name	**Rocket Propulsion**	**Course Code**	**AAE 4067**
Semester	**VII**	**Max Marks**	**50**
Date	**25/11/2024**	**Duration**	**3 hrs**

Instructions:
- **Answer ALL the questions.**
- **Data not provided may be suitably assumed**

Q. No.	Question	Marks	NBA		IET		BL
			CO	PO	CLO	AHEP4 LO	
1A	Explain the functioning of Ramjet and SCRamjet engines and assess their relevance and potential impact on high-speed flight technologies today.	2	CO5	1,2,3,4,6	CO5	1,12,18	4
1B	Compare the hybrid and solid propellant grains and regression rates.	3	CO4	1,2,3,4, 5,6,7,9,12	CO4	5,18	4
1C	Briefly explain the following terms: i) Progressive burning ii) Deflagration limit iii) Liner iv) Web thickness v) Case-bonded grains	5	CO3	1,2,3,4, 5,6,7,9,12	CO3	1,18	2
2A	A four stage rocket is used to insert a satellite of 40 kg mass in a Low Earth Orbit (LEO). The approximate values of mass of propellant, and structure, and jet velocity for each stage are given below. Determine: i) Payload mass fraction for the launch vehicle and mass ratio.	4	CO2	1,2,3,4, 5,6,7,9,12	CO2	1,2,3, 4,18	3

<table>
<tr><td></td><td>ii) Ideal velocity increment in each stage and the final burn-out velocity.

<table><tr><td>Stage</td><td>I stage</td><td>II stage</td><td>III stage</td><td>IV stage</td></tr><tr><td>Propellant mass (kg)</td><td>9000</td><td>3500</td><td>1700</td><td>260</td></tr><tr><td>Structural mass (kg)</td><td>1500</td><td>550</td><td>250</td><td>40</td></tr><tr><td>Jet velocity (m/s)</td><td>2200</td><td>2400</td><td>2500</td><td>2750</td></tr></table></td><td></td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td>2B</td><td>Write a brief note on the constituents of a solid propellant grain.</td><td>4</td><td>CO3</td><td>1,2,3,4, 5,6,7,9,12</td><td>CO3</td><td>1,18</td><td>2</td></tr>
<tr><td>2C</td><td>A solid propellant rocket producing 55 MN thrust is fired for 170 seconds. The specific impulse of the rocket is 2665 Ns/kg. Find the mass of propellant burned during the rocket operation and the effective exhaust velocity.</td><td>2</td><td>CO3</td><td>1,2,3,4, 5,6,7,9,12</td><td>CO3</td><td>1,18</td><td>3</td></tr>
<tr><td>3A</td><td>For a three-stage rocket with M_0 = 55,000 kg, and M_L= 4000 kg, the specific impulse of the stages is given as I_{s1} = 150 s, I_{s2} = 300 s, and I_{s3} = 350 s. The structural coefficients of the three stages are 0.13, 0.12, and 0.11, respectively. If the payload ratios of all three stages are the same, develop the equations for various masses and find the structural and propellant mass for the three stages. Compute the terminal velocity of the payload.</td><td>5</td><td>CO2</td><td>1,2,3,4, 5,6,7,9,12</td><td>CO2</td><td>1,2,3, 4,18</td><td>4</td></tr>
<tr><td>3B</td><td>Write a brief note on the gas generator rocket engine cycle.</td><td>3</td><td>CO2</td><td>1,2,3,4, 5,6,7,9,12</td><td>CO2</td><td>1,2,3, 4,18</td><td>2</td></tr>
<tr><td>3C</td><td>What are ablative materials? Explain how ablative materials help in cooling.</td><td>2</td><td>CO4</td><td>1,2,3,4, 5,6,7,9,12</td><td>CO4</td><td>5,18</td><td>2</td></tr>
<tr><td>4A</td><td>Sketch the performance plot of a convergent-divergent (C-D) nozzle and mark the critical points. Explain the behavior of the</td><td>5</td><td>CO1</td><td>1,2,3,4, 5,6,7,9,12</td><td>CO1</td><td>1,2,3,5, 12,18</td><td>3</td></tr>
</table>

	first, second, and upper stage rocket nozzles with schematics when tested at sea level conditions.						
4B	Describe the techniques used for reducing jet noise and infrared (IR) signatures in aircraft engines.	**3**	CO1	1,2,3,4, 5,6,7,9,12	CO1	1,2,3,5, 12,18	**3**
4C	Sketch the over-expanded C-D nozzle operation and the flow features.	**2**	CO1	1,2,3,4, 5,6,7,9,12	CO1	1,2,3,5, 12,18	**3**
5A	Briefly describe various types of subsonic combustion chambers used in aircraft engines. Discuss the widely used combustor in commercial aircraft.	**3**	CO5	1,2,3,4,6	CO5	1,12,18	**3**
5B	Describe the combustion phenomena in hybrid rockets.	**3**	CO4	1,2,3,4, 5,6,7,9,12	CO4	5,18	**3**
5C	Discuss the combustion chamber cooling process for aircraft and rocket engines.	**4**	CO5	1,2,3,4,6	CO5	1,12,18	**3**

MARK DISTRIBUTION (CO-wise & BTL-wise)

CO / CLO	Total Marks
1	***10***
2	***12***
3	***11***
4	***8***
5	***9***

BL	Total Marks
2	**14**
3	**26**
4	**10**
5	**0**